THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS CHAPTER TEST SET B

Chapter 3: **The World of Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The decimal expansion of $\frac{5}{6}$ is: 1

(a) non-terminating repeating **(b)** non-terminating non-repeating
(c) terminating **(d)** cannot be determined

2. The decimal expansion of $\frac{9}{14}$ is: 1

(a) terminating **(b)** non-terminating repeating
(c) non-terminating non-repeating **(d)** cannot be determined

3. The number $0.\overline{3}$ in the form $\frac{p}{q}$ is: 1

(a) 3 **(b)** $\frac{1}{3}$ **(c)** $\frac{2}{3}$ **(d)** $\frac{13}{30}$

4. The number $\sqrt{81}$ is: 1

(a) rational **(b)** irrational
(c) neither rational nor irrational **(d)** not a real number

5. The number π is: 1

(a) rational **(b)** not a real number
(c) neither rational nor irrational **(d)** irrational

6. A rational number in lowest terms has denominator 2×5^2. Its decimal expansion terminates after how many places? 1

(a) 2 **(b)** 4 **(c)** 1 **(d)** 3

7. The value of $0 - (-14)$ is: 1

(a) 14 **(b)** -14 **(c)** 18 **(d)** 16

8. The average of $\frac{1}{2}$ and $\frac{3}{4}$ is a rational number lying between them. It is: 1

(a) $\frac{5}{8}$ **(b)** $\frac{1}{4}$ **(c)** $\frac{7}{8}$ **(d)** $\frac{4}{5}$

9. Brahmagupta defined zero as the result of: 1

(a) dividing a number by itself
(b) multiplying by 1
(c) adding two equal numbers
(d) subtracting a number from itself ($a - a = 0$)

10. Between any two rational numbers there are: 1

(a) no rational number **(b)** infinitely many rational numbers
(c) exactly one rational number **(d)** exactly two rational numbers

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Without long division, state whether $\frac{3}{8}$ has a terminating or a repeating decimal. Give a reason. **2**

12. Without long division, state whether $\frac{7}{12}$ has a terminating or a repeating decimal. Give a reason. **2**

13. Without long division, state whether $\frac{11}{20}$ has a terminating or a repeating decimal. Give a reason. **2**

14. Without long division, state whether $\frac{5}{9}$ has a terminating or a repeating decimal. Give a reason. **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Express $2.45\overline{37}$ in the form $\frac{p}{q}$. **3**

16. Express $0.\overline{45}$ as a fraction in lowest terms and verify by dividing. **3**

17. Find three rational numbers between $\frac{1}{3}$ and $\frac{1}{2}$. **3**

OR

Find five rational numbers between 3 and 4. **3**

18. Find three rational numbers between 3.1415 and 3.1416. **3**

OR

Without long division state whether $\frac{77}{210}$ terminates. If it does, after how many places? **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A tailor has a roll of 15 m of silk; each kurta needs 2 m. Another roll has $12\frac{1}{2}$ m. **5**

(i) How many kurtas from the 15 m roll and how much silk is left?

(ii) How many from the second roll?

(iii) Express the silk left in each case as a rational number and find the total silk left.

OR

(i) Prove that $\sqrt{3}$ is irrational. **5**

(ii) Hence prove that $2 - \sqrt{3}$ is irrational.

20. Find 5 rational numbers between $\frac{3}{5}$ and $\frac{4}{5}$ by writing them with a common denominator large enough. Explain why the common denominator must be at least 6 times the difference of the numerators. **5**

OR

Case study: A square has side 1 unit. **5**

(i) Find the length of its diagonal.

(ii) Show how to represent it on the number line.

(iii) Why can this number not be written as $\frac{p}{q}$?

(iv) Write its decimal expansion to 4 places and say what kind of decimal it is.

END OF QUESTION PAPER — Total: 40 Marks